

FINANCIAL ANALYSIS REPORT

HBL Finance Analysis — Transactions, Customers & Risk

Power BI Dashboard Project

PREPARED BY

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REPORTING YEAR

2026 (Jan–Apr)

DATASET

HBL Finance Transactions & Customers Dataset (50,000 transactions / 5,000 customers)

REPORT TYPE

Transaction & Risk Performance Analysis

Executive Summary

This report analyzes HBL's transaction data for 2026 (January–April) to understand transaction volume, customer behavior, regional distribution, and operational risk. The analysis provides insights that can help improve transaction success rates, reduce risk exposure, and support better decision-making across customer segments and regions.

Business Overview

The bank processed 4,993 transactions in the period, generating a total transaction value of Rs 45.32M, of which Rs 38.93M (85.88%) came from successfully completed transactions. The average transaction size stood at Rs 9,077, with an overall failure rate of 9.75% by transaction count. Total fee income was Rs 75.44K and total tax collected was Rs 13.57K. Overall performance is stable, with clear opportunities to reduce failed transactions and grow underperforming regions and segments.

KPI Summary

KPI	Value
Total Transaction Amount	Rs 45.32M
Successfully Processed Amount	Rs 38.93M (85.88%)
Total Transactions	4,993
Average Transaction Value	Rs 9,077
Total Fee Income	Rs 75.44K
Total Tax Collected	Rs 13.57K
Failed Transaction Rate	9.75% (487 transactions)
Flagged Fraud Rate	1.28% (64 transactions)

Key Findings

- **Transaction Status** — Transaction Health: 85.88% of transaction value completed successfully; 10.28% failed and 3.84% remained pending.
- **Customer Segment** — Retail customers contribute Rs 24.08M (53%) of total transaction value, more than all other segments combined.
- **Regional Distribution** — Punjab leads with Rs 27.19M (60%), followed by Sindh at Rs 8.88M; Balochistan and ICT remain the smallest markets.
- **Gender Split** — Female customers account for Rs 23.39M (51.6%) versus Rs 21.93M (48.4%) from male customers — a fairly balanced split.
- **Transaction Type** — Loan EMI (Rs 12.31M) and Transfer (Rs 12.28M) are the two largest transaction types by value, together making up over half of total volume.
- **Risk Signal** — Loan EMI transactions alone account for Rs 1.57M of failed value — the single largest contributor to failures.

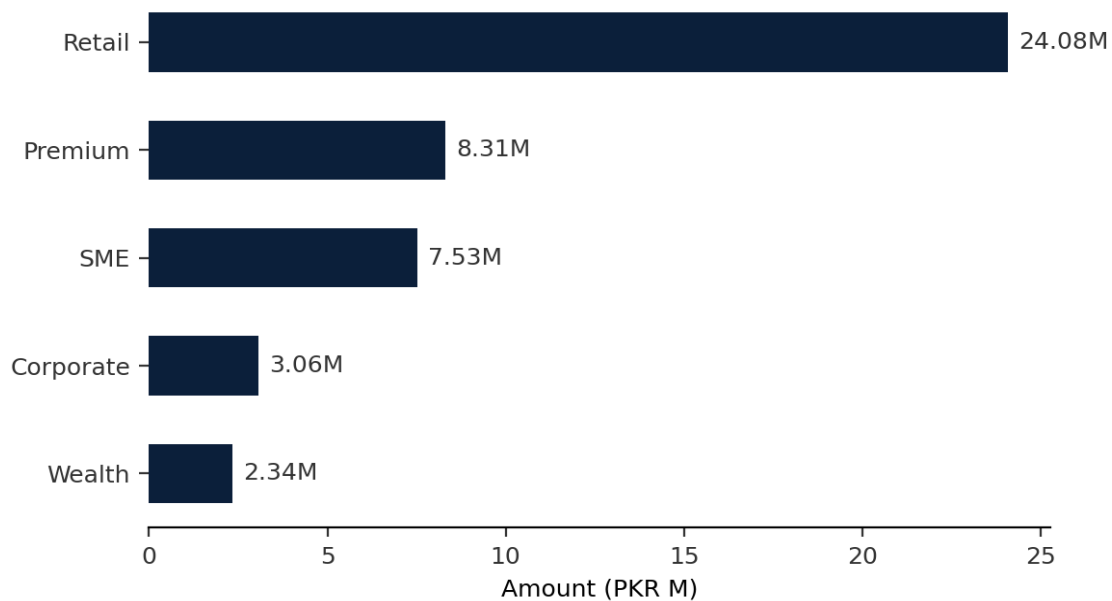
Detailed Business Analysis

Transaction Status Analysis

Out of Rs 45.32M in total transaction value, Rs 38.93M (85.88%) completed successfully, Rs 4.66M (10.28%) failed, and Rs 1.74M (3.84%) remained pending. By transaction count, 487 of 4,993 transactions (9.75%) failed — a rate worth addressing given its direct impact on customer trust and operational cost.

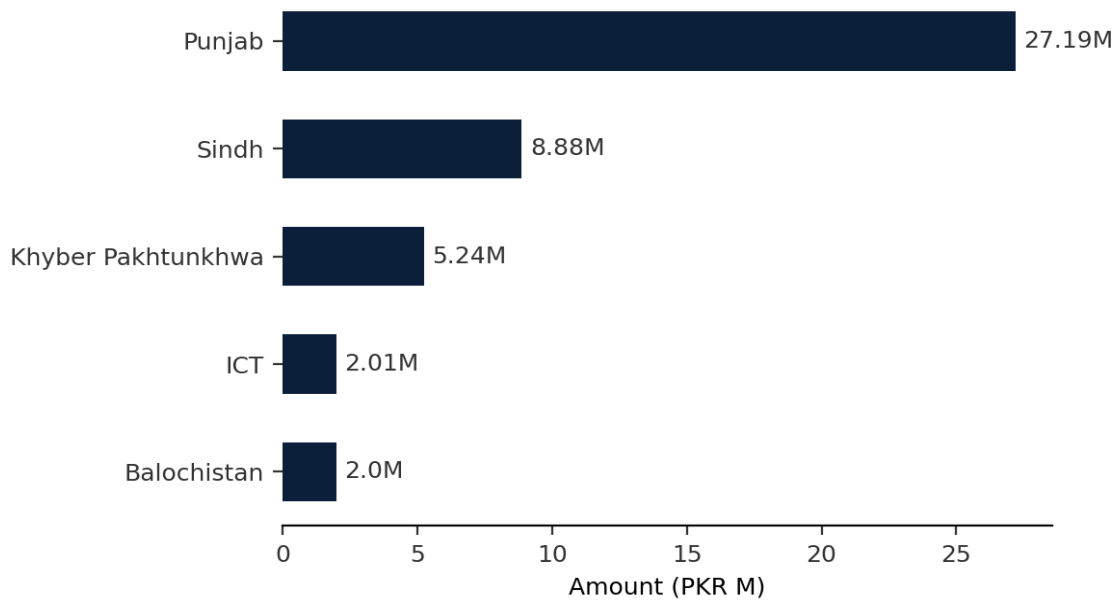
Failed transactions are concentrated in a few transaction types: Loan EMI (Rs 1.57M across 101 transactions) and Transfer (Rs 1.03M across 72 transactions) together account for over half of all failed value. By channel, Raast (Rs 1.34M failed) and ATM (Rs 0.91M failed) show the highest failure amounts, which may point to connectivity or timeout issues on these rails.

Customer Segment Analysis



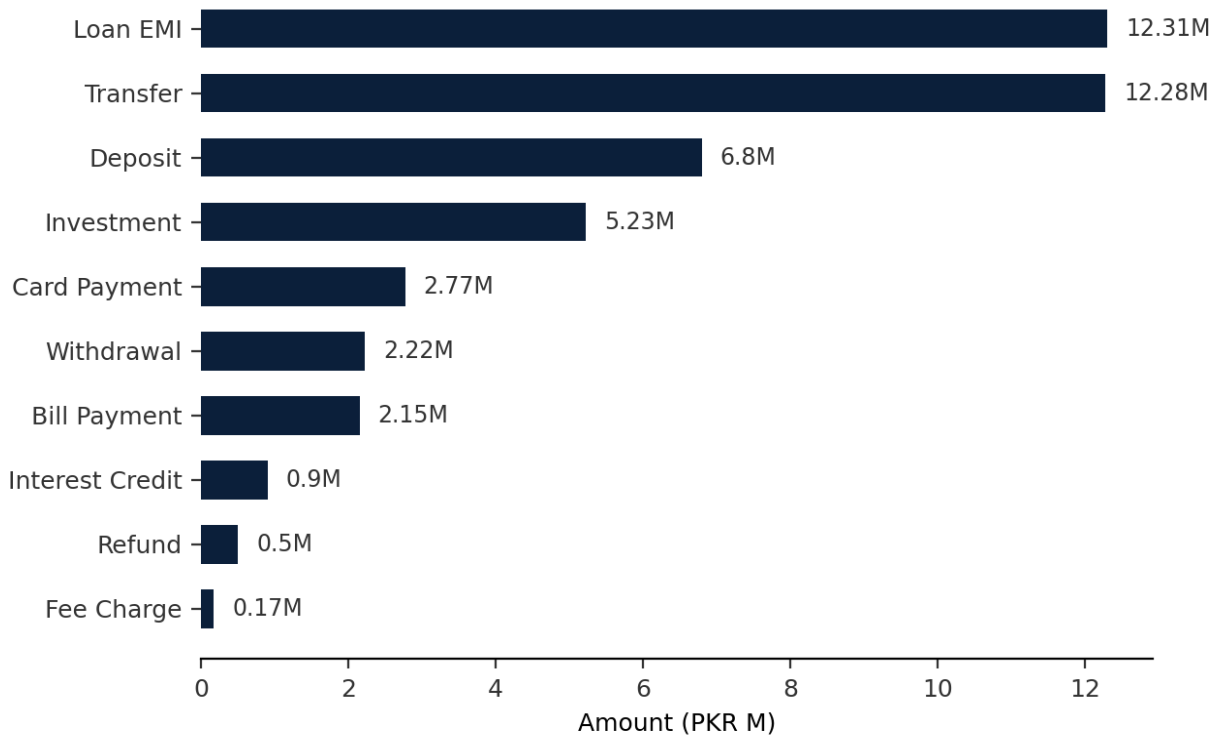
Retail is by far the dominant segment, contributing Rs 24.08M (53% of total value), followed by Premium (Rs 8.31M) and SME (Rs 7.53M). Corporate (Rs 3.06M) and Wealth (Rs 2.34M) contribute the smallest shares. This indicates the bank's transaction volume is currently driven almost entirely by everyday retail banking activity rather than high-value corporate or wealth relationships.

Regional Performance Analysis



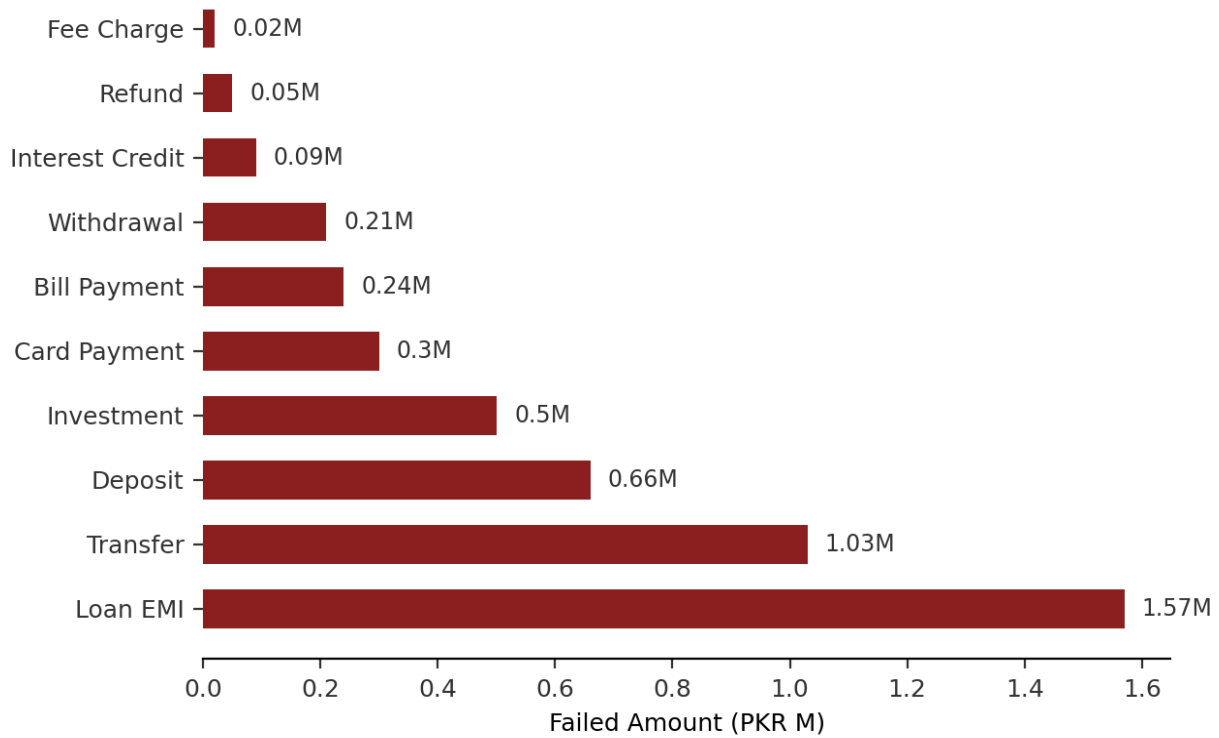
Punjab is the strongest-performing region at Rs 27.19M (60% of total value), followed by Sindh (Rs 8.88M) and Khyber Pakhtunkhwa (Rs 5.24M). ICT and Balochistan generate comparatively low transaction value (around Rs 2M each), suggesting a smaller customer base or lower branch/agent density in these regions.

Transaction Type Analysis



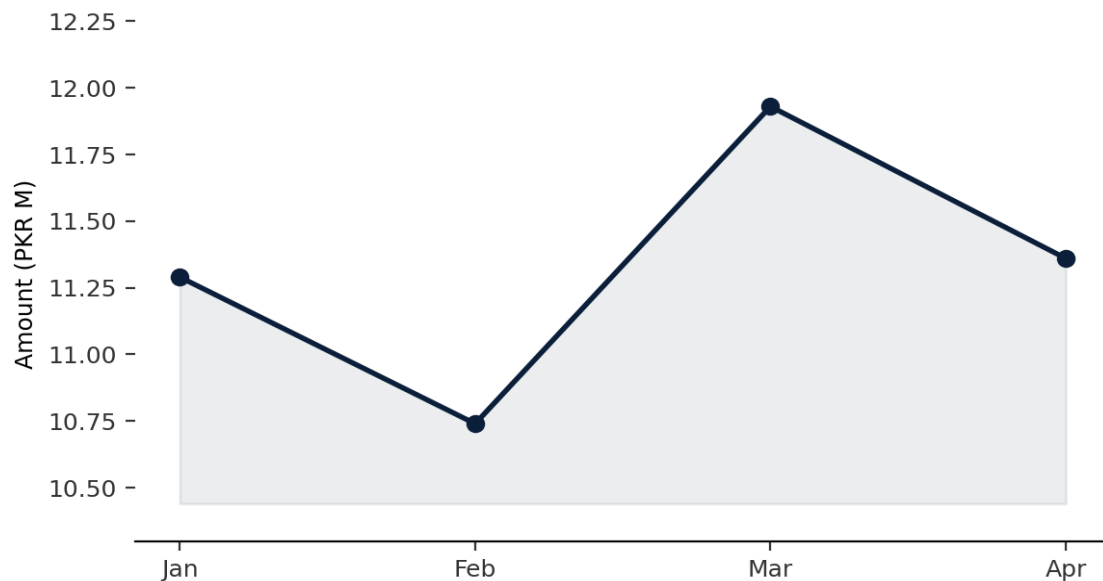
Loan EMI (Rs 12.31M) and Transfer (Rs 12.28M) are the two leading transaction types, together representing more than 54% of total value. Deposit (Rs 6.80M), Investment (Rs 5.23M) and Card Payment (Rs 2.77M) follow. Fee Charge and Refund transactions contribute the least, as expected for adjustment-type entries rather than core banking activity.

Failed Transactions by Type



Loan EMI transactions generate the highest failed value (Rs 1.57M), nearly 50% more than the next largest contributor, Transfer (Rs 1.03M). This concentration suggests the EMI auto-debit process may benefit from a closer review — for example insufficient balance retries, mandate expiry, or scheduling conflicts with salary credit dates.

Monthly Trend Analysis



Transaction value dipped in February (Rs 10.74M) before recovering to a high in March (Rs 11.93M), then easing slightly in April (Rs 11.36M). The overall pattern is relatively stable across the four months available, without a strong seasonal trend yet visible in this partial-year data.

Gender & Fraud Signal Analysis

Revenue is fairly balanced between female customers (Rs 23.39M, 51.6%) and male customers (Rs 21.93M, 48.4%), unlike segment or regional splits which are far more concentrated. On risk, 64 transactions (1.28% of volume, Rs 564.9K in value) were flagged as fraud — a low but non-zero rate that still warrants ongoing monitoring, particularly given flagged transactions carry a similar average value (Rs 8,827) to normal ones, making them harder to catch on amount alone.

Recommendations

1. Transaction Reliability

- Audit the Loan EMI auto-debit workflow — it drives the largest share of failed value; check retry logic, mandate validity, and timing against salary credit dates.
- Investigate Raast and ATM channel failures for connectivity, timeout, or integration issues, since these two channels carry the highest failed amounts.
- Set up automated retry and customer notification flows for failed Transfer and Deposit transactions to reduce repeat failures.

2. Customer Segment Strategy

- Design cross-sell campaigns to grow Corporate and Wealth segment activity, which currently contribute the smallest share of transaction value despite typically higher per-customer worth.
- Protect and deepen the Retail relationship with loyalty or cashback incentives, since it is the core revenue driver.

3. Regional Growth Strategy

- Increase branch/agent presence and marketing investment in Balochistan and ICT, the two lowest-performing regions.
- Study what is working in Punjab (channel mix, merchant partnerships) and replicate applicable elements in Sindh and KP.

4. Risk & Fraud Management

- Move beyond amount-based fraud rules, since flagged and normal transactions carry similar average values; incorporate velocity, location, and channel-based risk signals instead.
- Continue monitoring the risk_score field across transaction statuses, since Success transactions currently carry a marginally higher average risk score than Failed ones — worth validating the scoring model.

5. Reporting & Monitoring

- Extend the dashboard's date range as more of 2026 becomes available to track whether the Feb dip and Mar recovery repeat seasonally.
- Add a dedicated failed-transactions page filtered by type and channel so operations teams can act on the root causes identified above.

Business Impact

Implementing these recommendations is expected to:

- Reduce the 9.75% failed-transaction rate, improving customer trust and lowering reconciliation cost.
- Grow transaction value from underrepresented segments (Corporate, Wealth) and regions (Balochistan, ICT).
- Strengthen fraud detection accuracy beyond simple amount thresholds.
- Provide operations teams with a clear, filterable view of where and why transactions fail.
- Support more consistent transaction growth across regions and customer segments over time.

Dashboard Highlights

- Built a two-page interactive Power BI dashboard: an Overview Analysis page and a detailed Transactions page.
- Created 5 KPI cards (Total Amount, Total Transactions, Avg Transaction, Total Fee, Total Tax) each with a Year-over-Year comparison.
- Built a Dynamic Metric measure so a single line chart and set of cards can switch between Amount, Fee, Tax, and Transactions on user selection.
- Designed visuals for Transaction Status, Customer Segment, State, Gender, and Transaction Type, plus a detailed transaction-level table.
- Added slicers for Year, Dynamic Metric, Occupation, and Merchant Category for interactive filtering.
- Presented insights through a clean, HBL-branded navy theme with custom icons for each KPI card.

My Contribution to This Project

Beyond building the visuals, the bulk of this project's effort went into preparing clean, model-ready data in Power Query and building the DAX logic behind every KPI. This section documents that work.

A. Data Sources & Model

Two source tables were imported from CSV — finance_transactions (50,000 rows, 15 columns) and customers (5,000 rows, 10 columns) — connected through a customer_id relationship (many-to-one, single-direction filtering). A dedicated Calendar Table was created using CALENDAR() and marked as the official date table for time intelligence.

```
Calender Table = CALENDAR(
    MIN(finance_transactions[transaction_Date]),
    MAX(finance_transactions[transaction_Date])
)
```

B. Power Query — Data Cleaning Steps

finance_transactions table:

- Promoted headers and fixed inconsistent date formatting — replaced “-” separators with “/” and parsed transaction_date into a proper Date type (en-GB locale).
- Corrected column data types: amount, fee_amount, and tax_amount converted to Number; risk_score converted to Whole Number.
- Removed leading/trailing spaces from the channel column using Text.Trim.
- Removed duplicate rows based on transaction_id using Table.Distinct.
- Converted negative/inconsistent amount values to absolute values using Number.Abs, since a transaction amount should always be positive.
- Replaced null values in fee_amount with the column average (14.52) rather than dropping rows, to preserve transaction records for analysis.
- Reordered columns into a logical, readable sequence for the data model.

customers table:

- Fixed inconsistent date formats in both date_of_birth and join_date the same way as the transactions table.
- Merged first_name and second_name into a single customer_name column using Table.CombineColumns.
- Removed blank/unnamed placeholder columns left over from the raw CSV export.
- Reordered and renamed columns for a clean, presentation-ready schema.

C. DAX Measures

Every KPI card on the dashboard is backed by a base measure plus a matching prior-year and YoY% pair, using SAMEPERIODLASTYEAR against the Calendar Table:

```
Total Amount measure M = SUM(finance_transactions[amount])
previous year amount = CALCULATE([Total Amount measure M], SAMEPERIODLASTYEAR('Calender Table'[Date]))
YOY amount = [Total Amount measure M] - [previous year amount]
YOY % Amount = DIVIDE([YOY amount], [previous year amount])
```

The same four-measure pattern (base → previous year → YoY → YoY %) was repeated for Total Transactions, Average Transaction, Total Fee, and Total Tax — 12 additional measures in total. Other supporting measures:

```
Total Transactions = DISTINCTCOUNT(finance_transactions[transaction_id])
Avg Transaction = AVERAGE(finance_transactions[amount])
Total Fee measure = SUM(finance_transactions[fee_amount])
Total Tax measure = SUM(finance_transactions[tax_amount])
Selected year = SELECTEDVALUE('Calendar Table'[year])
```

Formatted string measures were also built so KPI cards display clean labels directly (e.g. “Rs 45.32M” instead of a raw number):

```
Total Amount = "Rs " & FORMAT([Total Amount measure M]/1000000, "0.00") & "M"
Total Fee = "Rs " & FORMAT([Total Fee measure]/1000, "0.00") & "K"
```

D. Dynamic Metric Switching

A calculated table (Dynamic Metric) was created using NAMEOF() to let users switch the line chart and cards between four measures from a single slicer, rather than building four separate charts:

```
Dynamic Metric = {
    ("Total Amount ", NAMEOF('finance_transactions'[Total Amount measure M]), 0),
    ("Total Fee", NAMEOF('finance_transactions'[Total Fee measure]), 1),
    ("Total Tax", NAMEOF('finance_transactions'[Total Tax measure]), 2),
    ("Total Transactions", NAMEOF('finance_transactions'[Total Transactions]), 3)
}
```

E. Visualization & Design

- Built KPI cards, a donut chart (Transaction Status), horizontal bar charts (Customer Segment, State), a matrix table (Transaction Type Analysis), and a ribbon/donut chart (Gender).
- Applied a consistent navy-and-white HBL brand theme with custom icons on each KPI card.
- Added Year, Dynamic Metric, Occupation, and Merchant Category slicers, and a second page for row-level transaction detail.

Conclusion

The HBL Finance Analysis dashboard shows a bank processing a healthy Rs 45.32M in transaction value over four months, with a clearly dominant Retail customer base and Punjab as the strongest region. The data cleaning work in Power Query — fixing dates, deduplicating records, handling nulls, and standardizing types — combined with a full YoY-enabled DAX measure framework, made it possible to build a dashboard that is both accurate and genuinely interactive through the dynamic metric switcher.

The clearest opportunity ahead is transaction reliability: Loan EMI failures alone account for the largest share of lost value, and the Raast and ATM channels show the highest failure amounts. Addressing these, alongside growing the underrepresented Corporate, Wealth, Balochistan, and ICT segments, would meaningfully strengthen both transaction success rates and overall portfolio balance.